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SELF-PREFERENCE BIAS IN LLM-AS-A-JUDGE

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ABSTRACT

Automated evaluation leveraging large language models (LLMs), commonly referred to as LLM evaluators or LLM-as-a-judge, has been widely used in measuring the performance of dialogue systems. However, the self-preference bias in LLMs has posed significant risks, including promoting specific styles or policies intrinsic to the LLMs. Despite the importance of this issue, there is a lack of established methods to measure the self-preference bias quantitatively, and its underlying causes are poorly understood. In this paper, we introduce a novel quantitative metric to measure the self-preference bias. Our experimental results demonstrate that GPT-4 exhibits a significant degree of self-preference bias. To explore the causes, we hypothesize that LLMs may favor outputs that are more familiar to them, as indicated by lower perplexity. We analyze the relationship between LLM evaluations and the perplexities of outputs. Our findings reveal that LLMs assign significantly higher evaluations to outputs with lower perplexity than human evaluators, regardless of whether the outputs were self-generated. This suggests that the essence of the bias lies in perplexity and that the self-preference bias exists because LLMs prefer texts more familiar to them.

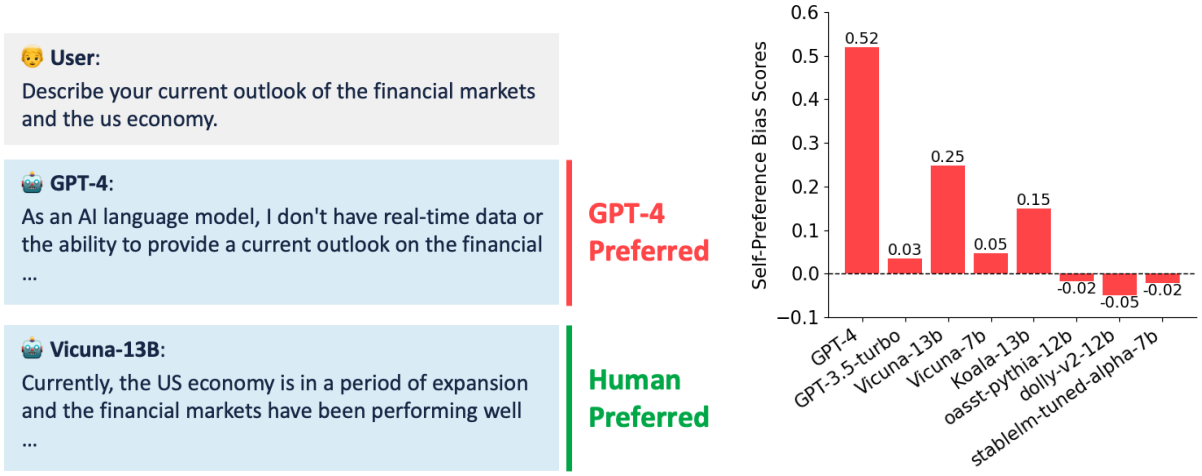
Keywords LLM-as-a-Judge · Self-preference bias · Fairness

1 Introduction

Measuring the quality of responses in dialogue systems presents unique challenges due to the diverse range of response strategies. Nowadays, thanks to the strong text understanding of large language models (LLMs), various automatic evaluations can be easily implemented by employing LLMs as evaluators (often referred to as LLM evaluators or LLM-as-a-judge). MT-Bench [Zheng et al., 2024] is an example of a benchmark utilizing this approach to score dialogue systems.

However, as Deutsch et al. [2022] reported, LLM evaluators are inherently biased. The inherent bias might cause inappropriate testing and inject biased preferences into the target dialogue systems [Zheng et al., 2024]. One of the most significant biases is self-preference bias, which refers to the tendency of LLMs to overestimate the quality of their own outputs, as demonstrated in Figure 1a. The self-preference bias poses potential risks, including the promotion of specific ideologies or response styles intrinsic to the LLM evaluator.

Several studies have addressed the issue of self-preference bias; however, there is a lack of reliable metrics to quantify the extent of self-preference bias, and the fundamental causes of this phenomenon remain unclear. Panickssery et al. [2024] reported on the relationship between self-preference bias and self-recognition ability, but their work lacked a comparison with human evaluators. Xu et al. [2024] and Stureborg et al. [2024] addressed quantifying self-preference bias within an evaluation approach where LLMs assign an absolute score to a single generated text. In this approach, gold-standard evaluators are required to assign scores based on abstract criteria while ensuring consistency with prior evaluations, making it challenging to obtain accurate assessments. As a result, these studies often restricted their



(a) (b)

Figure 1: **How much do LLMs prefer their own responses over human evaluations?** (a) illustrates an example where GPT-4 favors its own response, even when human evaluations prefer to a response generated by Vicuna-13B. (b) compares the self-preference bias scores using our proposed metric (Definition 4.1). These figures demonstrate that GPT-4 exhibits a stronger self-preference bias than other models, suggesting that it tends to rate its own outputs more favorably than human evaluations. For detailed experimental settings, refer to Section 4.

scope to specific tasks, such as text summarization or machine translation, and relied on reference-based metrics like BLEURT [Sellam et al., 2020], which does not reflect the diversity of real-world use cases.

By contrast, a pairwise evaluation approach that involves direct comparison between two texts enables evaluators to recognize specific differences more readily, resulting in more consistent human judgments. Consequently, such pairwise evaluation methods are particularly suitable for analyzing biases related to discrepancies with human evaluations.

In this paper, we measure the self-preference biases of LLMs in the pairwise evaluation. To accomplish this, we propose a new metric to quantify self-preference bias on the basis of algorithmic fairness concepts, thereby enabling discussions within the existing frameworks of fairness. In our experiment, we measured self-preference bias in eight LLMs. The results indicated that GPT-4 exhibited a significant self-preference bias (Figure 1b). This finding suggests a potential concern: using GPT-4 as a judge may lead to excessive influence from GPT-4's unique styles and policies.

Furthermore, we investigated the underlying causes of self-preference bias. Although LLM evaluators are not explicitly informed whether a given text is their own, they still exhibit self-preference bias. We hypothesized that LLM evaluators might be affected by the perplexity of the text, which tends to be lower perplexity when it is generated by themselves. To test this hypothesis, we analyzed the relationship between the perplexities of the texts to be evaluated and their corresponding evaluations. Our analysis revealed that LLMs assign significantly higher evaluations to texts with lower perplexity than human evaluators, regardless of whether the texts were self-generated. This suggests that the fundamental cause of self-preference bias may be the familiarity of the texts to the LLM evaluators, specifically how likely they are to generate the same response.

Contributions The contributions of this paper are threefold: (1) We propose a new metric to quantify self-preference bias in LLMs; (2) Using this metric, we evaluate the extent of self-preference bias across eight different LLMs; and (3) We identify a tendency for LLMs to assign higher ratings to texts with lower perplexity while exploring potential causes for this self-preference bias.

2 Related Work

LLM-as-a-judge has been studied in various directions. The use of large language models (LLMs) for benchmarking has been advancing steadily. LLMs are increasingly employed in dialogue evaluation due to their ability to provide flexible assessments from perspectives such as utility and safety [Zheng et al., 2024, Sottana et al., 2023, Wang et al., 2024, Schick et al., 2021]. Furthermore, numerous studies have explored the use of feedback from LLMs to enhance

the learning of the LLMs themselves [Madaan et al., 2023, Chen et al., 2024]. In particular, pairs of texts annotated with their relative quality have been proven to be useful training signals [Sun et al., 2023, Yuan et al., 2024], combined with various learning methods [Ouyang et al., 2022, Rafailov et al., 2023].

The self-preference bias is only one of several limitations inherent in the LLM-as-a-judge paradigm. Zheng et al. [2024] identified other biases such as position bias, where specific positions within a prompt are preferentially selected, and verbosity bias, which favors longer responses. Position bias can be addressed relatively easily by rearranging the order of comparative options within the prompt, and alignment-based methods to mitigate the bias have also been proposed [Li et al., 2024]. Additionally, Saito et al. [2023] investigated the presence of verbosity bias in GPT-4 and GPT-3.5-Turbo, focusing on the quantification of bias by examining the error rates associated with longer versus shorter text options.

3 Preliminaries: Fairness and Bias

The measurement of bias in classifiers has been widely discussed within the framework of fairness [Calders et al., 2009, Hardt et al., 2016]. In this section, we focus on a representative definition of fairness, Equal Opportunity, as the preparation for quantifying the self-preference bias in LLM evaluators.

Equal Opportunity [Hardt et al., 2016] is a fairness definition that requires the classifier to achieve equal recall across groups with different sensitive attributes (e.g. gender or race). Specifically, let the sensitive attribute be $S \in \{0, 1\}$, the prediction of the classifier be $Y' \in \{0, 1\}$, and the ground truth label be $Y \in \{0, 1\}$. The classifier satisfies Equal Opportunity if the following condition holds:

$$P(Y' = 1|S = 1, Y = 1) = P(Y' = 1|S = 0, Y = 1) \tag{1}$$

When quantifying bias in the classifier, the difference or ratio between both sides of the equation are often used. The amount of bias derived from this definition is the difference in how well the classifier matches the ground truth between groups with different sensitive attributes. Therefore, if there is already bias towards sensitive attributes during the creation of the ground truth, Equation 1 cannot be considered an ideal definition of fairness.

To address this drawback, the fairness definition known as Demographic Parity [Calders et al., 2009], which does not rely on ground truth, is also widely used. Demographic Parity requires that the predictive distribution of the classifier be consistent across groups with different sensitive attributes. It is based on the assumption that there is no inherent causal relationship between the sensitive attributes and the predicted labels.

LLM-as-a-judge is a technique aimed at replacing human evaluators with LLMs, and more specifically, it assumes that dialogue system are aligned based on human preferences. Therefore, in this study, we employ the concept of Equal Opportunity to quantify self-preference bias by treating LLM evaluators as classifiers.

4 Quantifying Self-Preference Bias

In this section, we propose a new metric to quantify self-preference bias. In particular, we focus on a setting where evaluators compare two texts and select the one higher quality, allowing comparison with reliable human evaluations.

4.1 Self-preference Bias Metric

To measure the extent to which an LLM's evaluation deviates from human evaluations across its own responses and those generated by others, we employ the concept of Equal Opportunity [Hardt et al., 2016]. The bias is quantified by calculating the difference between both sides of Equation 1. A rigorous definition is provided below.

Definition 4.1. *(self-preference bias of the evaluator f.) Let f be the evaluator assessing the quality of dialogue responses, capable of generating its own responses. For a pair of responses y_0 and y_1 in a dialogue, define: $Y \in 0, 1$ as the index of the human-preferred response, $Y' \in 0, 1$ as the index of the f-preferred response, and $S \in 0, 1$ as the index of the f-generated response. We define the self-preference bias of the evaluator f in dialogue comparison evaluation as follows:*

$$\textit{Bias} = P(Y' = 1|S = 1, Y = 1) - P(Y' = 1|S = 0, Y = 1). \tag{2}$$

In this definition, the amount of bias is represented by the difference between the conditional probability of the evaluator f rating itself favorably given that the human evaluator has rated it favorably, and the conditional probability of the evaluator f rating itself unfavorably given that the human evaluator has rated it unfavorably. A value of 0 indicates the